Trader Performance vs Market Sentiment Analysis

Objective

The objective of this analysis is to understand how market sentiment (Fear vs Greed) impacts trader performance and behavior.

Data Preparation

Two datasets were used:

- Market sentiment dataset (Fear/Greed)
- Trader transaction dataset

Steps performed:

- Cleaned data and checked missing values (none found)
- Converted timestamps to date format
- Merged both datasets using date

Key Metrics

- Daily Profit and Loss (PnL)
- Win Rate (percentage of profitable trades)
- Trade Frequency
- Average Trade Size

Analysis

Performance vs Sentiment

- Extreme Greed shows highest average PnL
- Win rate is also highest during Extreme Greed
- Fear and Extreme Fear show lower performance

Behavior Analysis

- Trading activity varies with sentiment
- Traders take more actions during emotional market phases

Segmentation

- Frequent traders execute more trades
- Infrequent traders show limited activity

Key Insights

- Traders perform best during Extreme Greed market conditions
- Market Fear negatively impacts trading performance
- Trader behavior changes significantly based on sentiment

Strategy Recommendations

- Reduce trading activity during Fear periods to avoid losses
- Apply risk control during Extreme Greed to avoid overconfidence

Conclusion

Market sentiment has a strong influence on trader behavior and performance. Understanding these patterns can help design better trading strategies.